Otoli

The science of independent sleep.

Otoli is a smart infant sleep system engineered from the vestibular system out: a multi-year platform that adapts to each infant in real time, and is designed to make itself unnecessary. Developed by Vestibrum, Inc.

01 / THE PROBLEM

The category sells a patch. Otoli was built to *end the problem*.

New parents lose an average of 44 days of sleep in the first year. Category leaders such as SNOO have already demonstrated that parents will pay a premium for infant sleep technology.

But every product in the category shares the same three limits: it **expires at six months**, it **reacts to crying** rather than building a path toward independent sleep, and it **returns the family to the original problem** at the crib transition. Parents buy a device that works, then age out of it and start over.

02 / THE SOLUTION

A multi-year platform engineered from the *vestibular system* out.

Infant sleep is not luck or habit. It is triggered by the vestibular system of the inner ear, and Otoli is the first sleep system that learns each baby's response to it and adapts in real time.

01 · Adaptive, not preset.

While the baby sleeps, Otoli reads movement, breathing, and sound, continuously tuning calibrated motion and acoustic cues to what is actually working, learning each infant's patterns night over night.

02 · One purchase, three years.

Bassinet → crib → toddler bed on a single smart base. Roughly two years of additional usable life compared with the premium products it replaces.

03 · Designed to make itself unnecessary.

A closed-loop weaning protocol gradually reduces stimulation as the child develops self-soothing capacity. The outcome is independent sleep, which no competitor offers.

03 / THE SCIENCE

The mechanism is *otolith-mediated*, and the intelligence is closed-loop.

The infant vestibular system contains two otolith organs, the saccule and the utricle, that sense **linear acceleration**. Otoli's defining feature is not the motion that engages them, but the closed-loop intelligence that governs it.

As the infant sleeps, the system continuously senses movement, breathing, and sound, and adapts its calibrated linear motion and acoustic cues in real time. This differs fundamentally from the fixed rotational rocking of motion-based competitors, which run preset patterns and primarily engage the semicircular canals.

> *In adult mice, the sleep-promoting effect of gentle lateral motion depended on functional otolith organs.*
>
> KOMPOTIS ET AL., CURRENT BIOLOGY (2019); SEE ALSO PERRAULT ET AL., CURRENT BIOLOGY (2019)

Most importantly, the same intelligence that soothes also weans. As the baby develops the capacity to self-soothe, Otoli imperceptibly withdraws its assistance: working invisibly, personalizing continuously, ultimately designed to make itself unnecessary.

04 / THE MARKET

A premium segment, in one of wellness's *fastest-growing* categories.

$1.4B	~500K	$3.5B
U.S. SMART-SLEEP TAM	PREMIUM BIRTHS PER YEAR	GLOBAL BABY SLEEP PRODUCT SALES

Sleep wellness is one of the fastest-growing segments of the global wellness economy. Parents are increasingly investing in premium connected wellness products, while advances in sensor technology and adaptive AI now make intelligent infant sleep systems commercially viable for the first time.

Otoli targets the premium child sleep market at a projected **$1,899 MSRP**, matching the category price point, with roughly two years of additional usable life. The platform extends through add-on crib and toddler modules, replacement mattresses, and a premium app subscription. B2B specialty channels (luxury hospitality and hospital/NICU programs) open in Year 2 and beyond.

Where competitors' customer relationships end at six months, Otoli's compound for three years.

05 / DEFENSIBILITY

The weaning protocol is the *core of the IP*.

Vestibrum's novelty search confirmed that **no prior IP performs longitudinal, multi-week weaning toward independent sleep**. Combined with the patent-pending three-axis linear-motion platform and multimodal sensing, it forms an integrated system with no competitor equivalent. It is the source of both Otoli's efficacy and its patent position.

PRODUCT	LIFESPAN	MOTION	PATH TO INDEPENDENT SLEEP
SNOO	6 mo	Single-axis rock + vibration	None; reactive cry escalation
Cradlewise	24 mo	Single-axis vertical bounce	None; camera-based sleep learning
Maxi-Cosi / Elvie / HALO	6 mo	1–2-axis rock or bounce	None; cry detection and response
Bosch Revol	24 mo	Minimal (single-axis vertical)	None; monitoring-led (camera + radar)
Otoli	**36+ mo**	**Three-axis linear system**	**Closed-loop weaning protocol**

06 / TRACTION

De-risked before the *first dollar*.

INTELLECTUAL PROPERTY

USPTO provisional patent filed February 2026. Novelty search complete, confirming the closed-loop weaning protocol as the lead claim.

ENGINEERING

A leading product-engineering firm is engaged to deliver a functioning prototype within nine months.

CORPORATE

IP, brand, and team secured under a Delaware C-Corporation.

07 / ROADMAP

Three milestones *reprice the company*.

KICKOFF · MO. 2	PROTOTYPE · MO. 9	PILOT · MO. 12	KICKSTARTER · MO. 14
SEED · MO. 18	SCALE · MO. 24		

Working hardware, infant pilot data, and demonstrated consumer demand, in that order.

08 / THE TEAM

Clinical depth, paired with *operating scale*.

Aalap C. Shah, MD

FOUNDER & CEO

Board-certified pediatric anesthesiologist with over a decade of clinical practice and originator of the Otoli mechanism. Direct domain expertise in infant physiology and pediatric device design.

Kaiya Bercow

CO-FOUNDER & COO

Operator across multiple consumer DTC brands, including a successful exit. Leads commercialization, brand, and go-to-market.

Jonathan Dodd

CHIEF FINANCIAL OFFICER

Mechanical engineer and ICAEW Fellow with public-market financing and capital-raise leadership, and the engineering fluency to hold technical partners accountable.

Khyati Davey

ADVISOR

Fractional MedTech executive with commercialization experience across regulated medical devices.

09 / VISION

Transform one of the most exhausting periods of early parenthood into a *restorative* one.

Safer, smarter, and more restorative, for every family, not just the first six months.

NEXT STEP

Let's find *thirty minutes*.

Aalap takes investor conversations directly. Pick a time that works. No gatekeeping, no forms.

SCHEDULE WITH AALAP · calendly.com/shlaapy/30min

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